Investigating GNSS-Reflectometry on Antarctic Glacial Surfaces

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Background

GNSS Reflectometry (GNSS-R) is a remote sensing technique that uses GNSS satellite signals after reflection from the Earth's surface. In this work, a downward-facing antenna was directed toward the ground to capture signals reflected from the glacial surface. These reflected signals are influenced by surface properties such as roughness, snow cover, exposed ice, and meltwater, making GNSS-R a potential tool for monitoring surface conditions. This is especially valuable in Antarctica, where large regions are difficult to observe directly and changes in snow and ice are important to monitor.

Data was collected on the McMurdo Ice Shelf at two sites: Phoenix (PHNX), which was primarily snow-covered, and Pegasus (PGSS), which contained mixed bare ice and patchy snow. The campaign included 6 observation dates at PHNX and 7 at PGSS between November and December 2023, producing 62 total observation instances and 33 significant collection periods longer than one minute, for approximately 2.25 TB of data. **This work examines whether signals reflected from glacial surfaces can be used to identify and study surface conditions.**

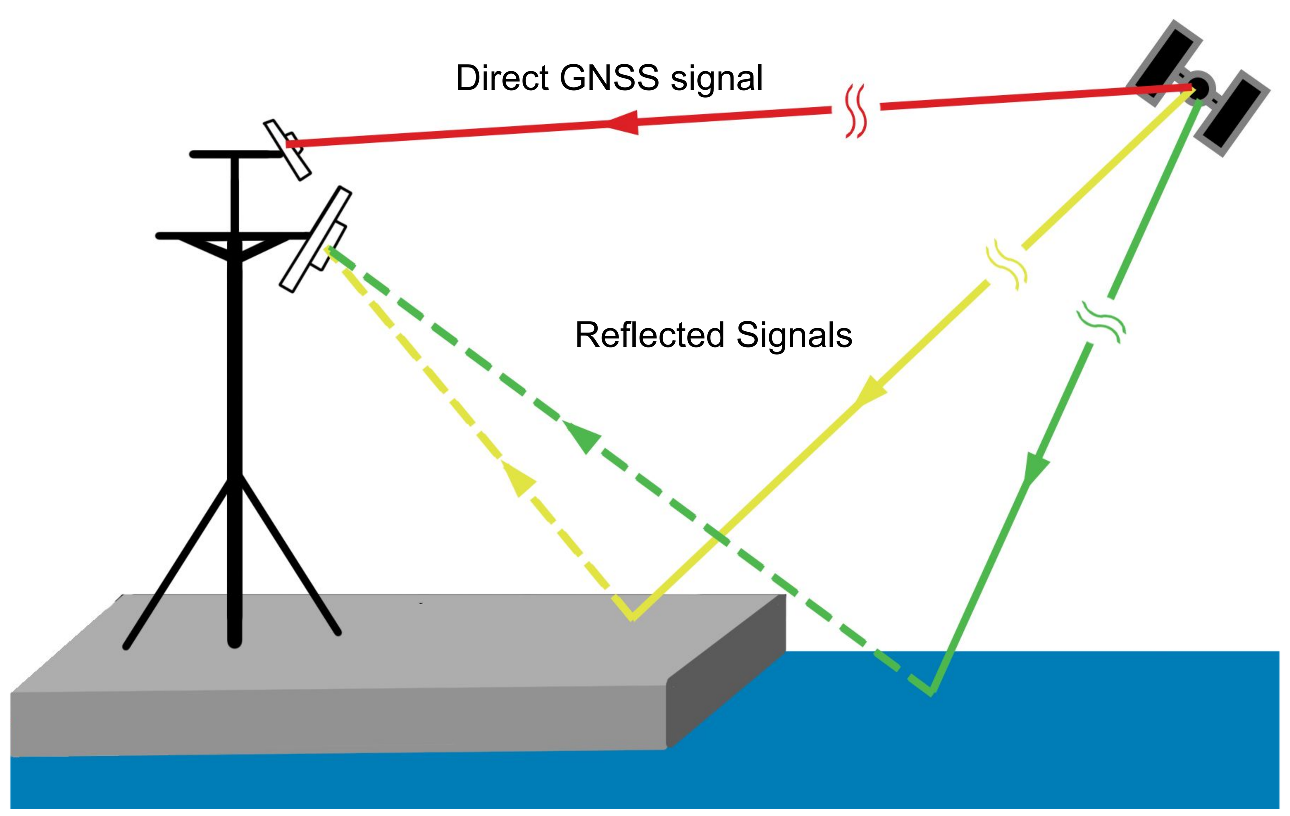

Figure 1. Illustration of the GNSS-R concept. A downward pointing antenna receives the GNSS signal after it has interacted with the surface being sensed.

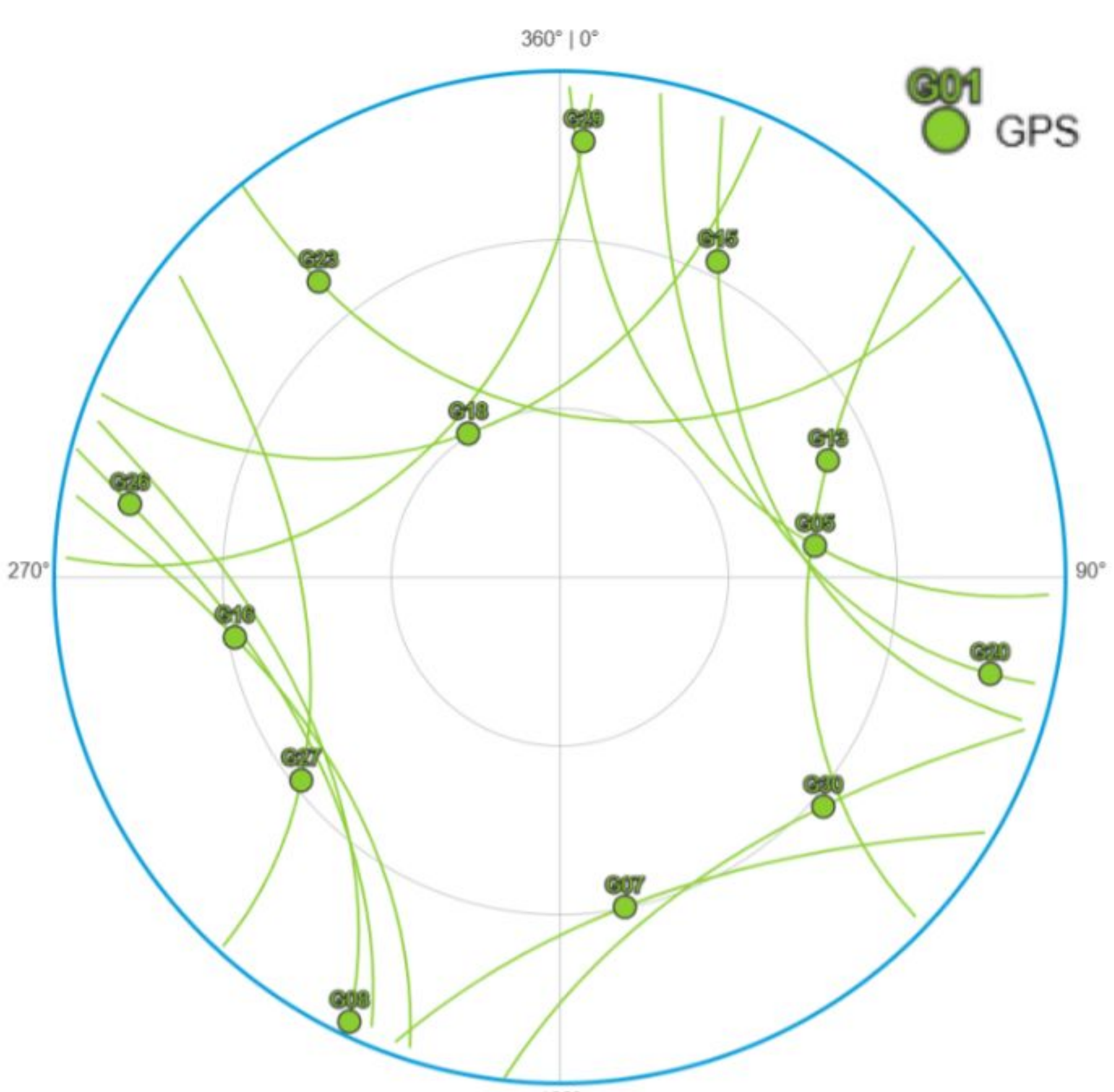

Figure 2. Skyplot of satellites (PRNs) in the sky at the PGSS site on December 8th, 2023 at 03:40 UTC, the time of one of the data recordings. These PRNs are expected to have a higher correlation signal than PRNs below the horizon.

Skyplot generated using Trimble GNSS Planning Online.

Data Verification

Raw binary data was first inspected to confirm that usable GNSS-R measurements had been collected. Time-domain traces, frequency-domain power spectral density estimates, and sample-value histograms were used as quick checks of signal behavior and spectral content. This step helped separate usable observations from questionable collections and identify noise-only cases. In particular, the first five PHNX measurements from December 8, 2023 were excluded from later analysis because their spectral behavior and field notes indicated that the USRP was not properly connected and was likely collecting only noise.

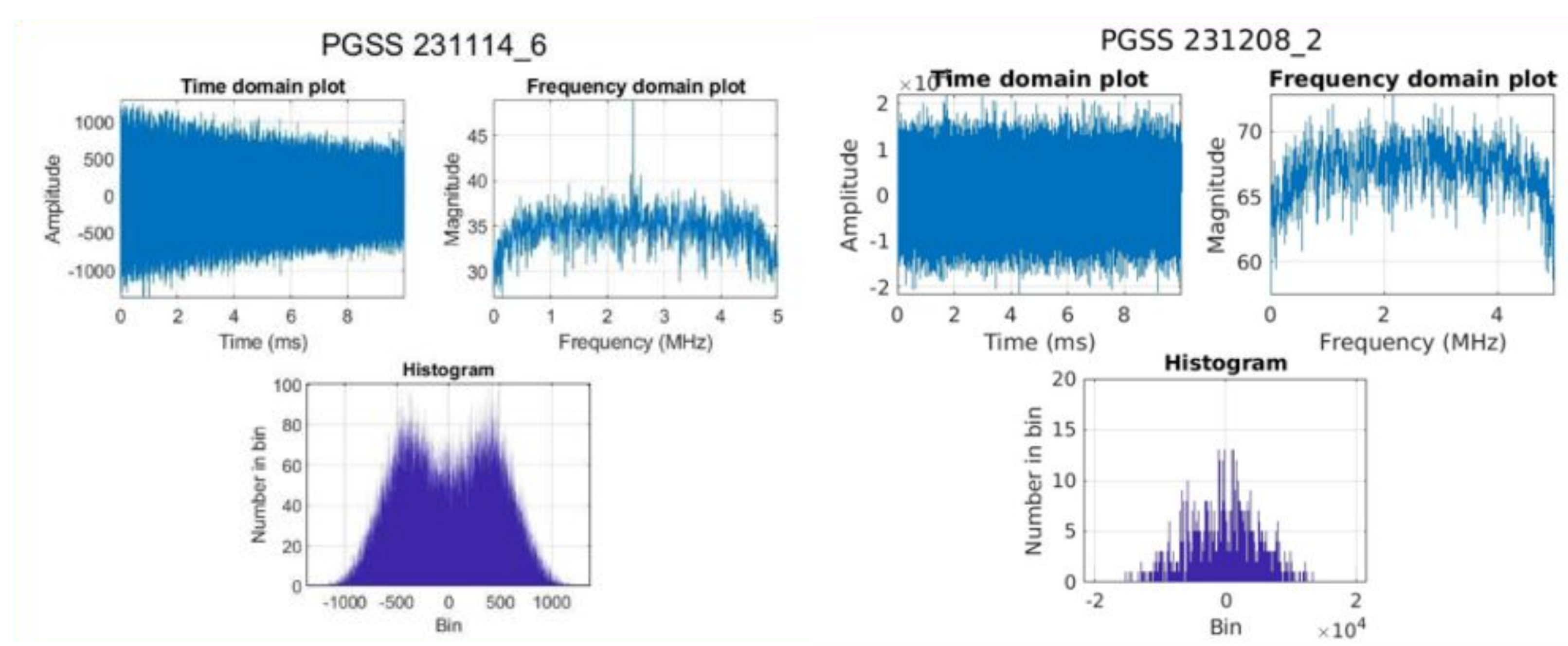

Figure 3. Testing plots of the raw, unprocessed data files. Left: pure-noise calibration collected before the field campaign. Right: data collected on December 8, 2023. The two cases differ in their time-domain behavior, frequency-domain shape, and histogram distribution. These comparisons were used as an initial check to distinguish noise-only input from usable collected data.

Signal Processing

After verification, the reflected GNSS data were processed to search for weak satellite signals using signal-to-noise ratio (SNR) metrics. The data was correlated with the expected satellite code over short coherent integration intervals, then combined incoherently over longer times to strengthen persistent signal energy. For this analysis, the data was grouped into 10 ms coherent blocks. Adjacent blocks were compared in pairs, and the stronger block from each pair was retained for the final sum. This reduced the likelihood of bit flips degrading the computed SNR.

Signal strength was evaluated using a metric labeled SNR2. In the acquisition script, SNR2 was defined as the ratio of the strongest correlation peak above the mean background level to that mean background level, as shown in Equation 1:

$$SNR2 = \frac{peakSize - Exp\_v}{Exp\_v} \tag{1}$$

Here, the noise term was not taken from the full signal, but from the mean background level surrounding the primary peak. This differs from the more traditional acquisition metric used in prior studies, which is based on the ratio of the primary peak to the secondary peak. This distinction is important for reflected GNSS signals, since surface scattering can produce competing secondary peaks; using local peak-to-background measures can therefore be more informative for weak reflected signal detection.

Results

- SNR2 distributions were compared across satellites, labeled by PRN number, to determine whether satellites expected to be visible in the sky (Figure 2) behaved differently from those not expected to be visible.
- Analysis was limited to the December 8, 2023 Part 2 dataset, one of six datasets collected on that date.
- In-sky PRNs generally showed occasional larger SNR2 spikes and higher maximum SNR2 values than out-of-sky PRNs.
- This behavior was most apparent for PRNs with clearer reflection geometry and less interference from nearby multipath sources, such as the secondary tower.
- Overall, the distributions did not separate clearly enough to distinguish visible from non-visible cases.
- Figure 4 shows no clear overall separation in the combined histogram of all 32 PRNs. Figure 5 shows some clustering between mean and maximum SNR2, but no strong distinction because outliers remain.

These results suggest that some visible satellites may produce stronger reflected signals useful for further study, but the current processing method does not yet definitively acquire signals or extract enough information for reliable surface analysis.

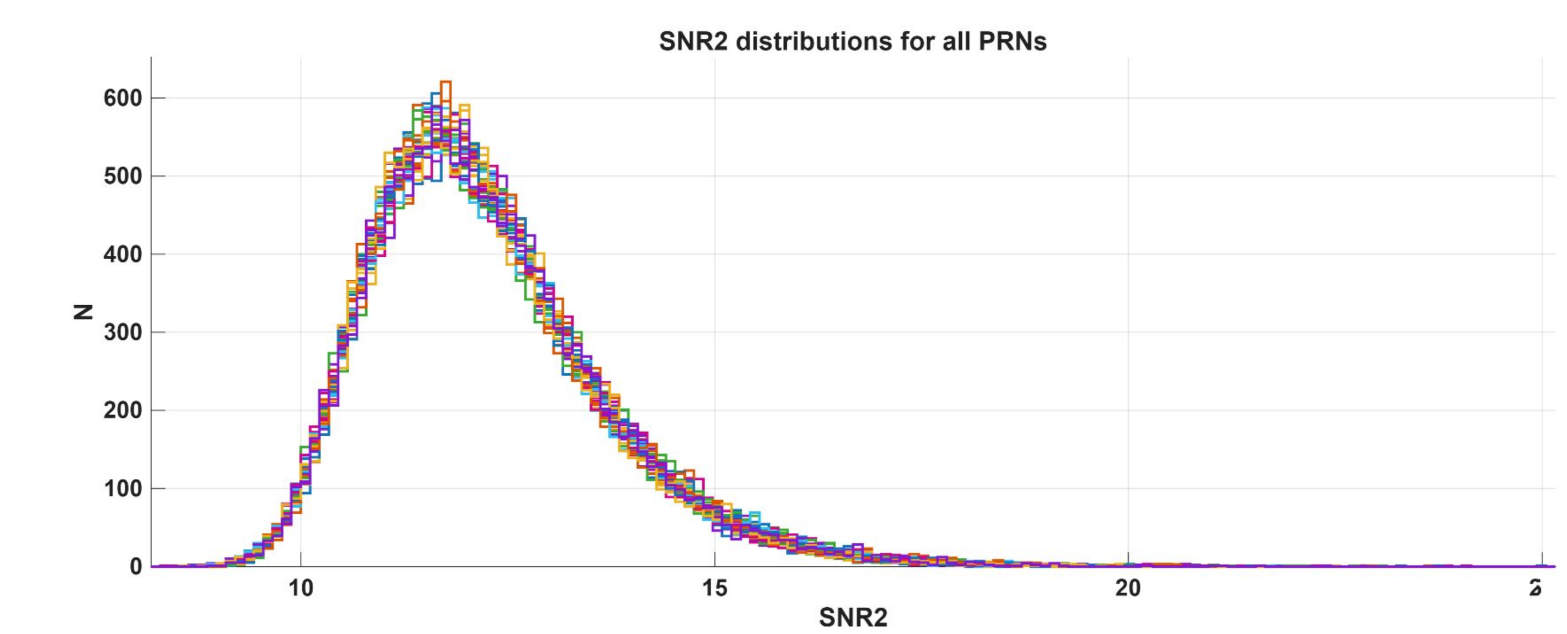

Figure 4. Combined SNR2 distributions for all 32 PRNs in the December 8, 2023 Part 2 dataset. The distributions largely overlap and show similar overall shapes, indicating no clear separation between PRNs based on SNR2 distribution alone.

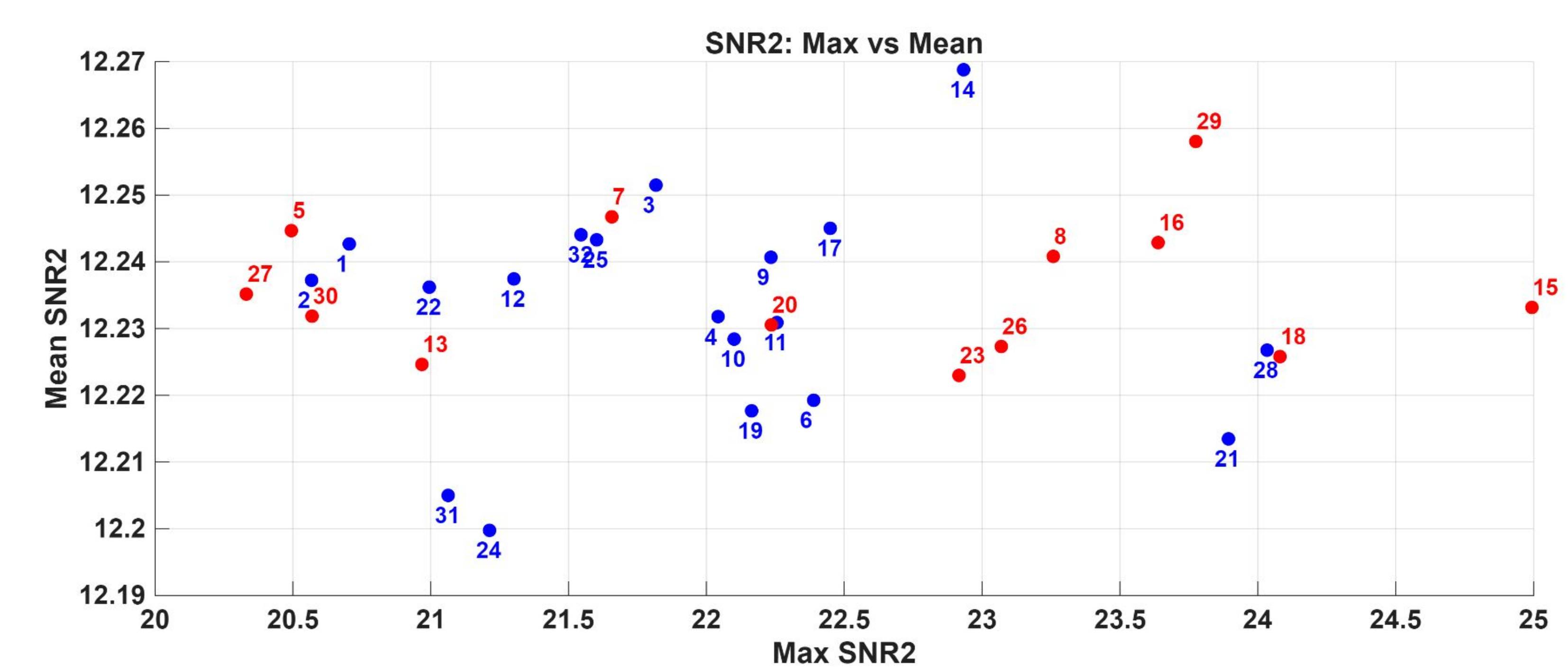

Figure 5. Mean SNR2 versus maximum SNR2 for all 32 PRNs in the December 8, 2023 Part 2 dataset. Red labels indicate in-sky PRNs and Blue labels indicate out-of-sky PRNs. Some partial separation is visible, with several in-sky PRNs reaching higher maximum SNR2 values, but overlap and outliers remain.

Future Work & Acknowledgements

- Expand the in-sky vs out-of-sky comparison across additional dates
- Improve SNR processing and reflected-signal detection
- Compare insights from the GNSS-R and GNSS-IR analyses

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